Inference Is Learning: No Phase Separation in River Learning

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Abstract

Modern deep learning draws a strict boundary between training and inference. During training, a forward pass is followed by a backward pass and a weight update; during inference, weights are frozen and only the forward pass is executed. This separation is accepted as a foundational assumption, yet it has no theoretical necessity.

We show that River Learning [1] operates without this separation. A single `carve()` operation simultaneously performs prediction (inference) and resistance update (learning). Three properties emerge from this structural fact alone:

1. **Unsupervised learning**: semantic structure is discovered without any labels.
2. **Interpretability**: each node directly explains what it learned.
3. **Continual learning**: weights update during inference, with no mode switch.

In an unsupervised word representation experiment at Word2Vec scale: WS353 Spearman correlation of 0.2576 is achieved (vs. Word2Vec dim=300 $\approx$ 0.30), with 1.6M parameters (27% of Word2Vec dim=300), no negative sampling, no labels, and no gradient. Eighty highway nodes self-organize into interpretable semantic clusters without any supervision.

1 Introduction

The standard deep learning training loop is:

$$\text{train}(x, y): \quad x \xrightarrow{\text{fwd}} \hat{y} \xrightarrow{\text{loss}} \mathcal{L} \xrightarrow{\text{bwd}} \nabla\mathcal{L} \xrightarrow{\text{update}} \theta$$

$$\text{infer}(x): \quad x \xrightarrow{\text{fwd}} \hat{y} \qquad [\theta \text{ frozen}]$$

This two-phase structure is so pervasive that it is rarely questioned. Yet the brain does not work this way. Synaptic connections strengthen during the very act of processing new experience; learning and inference are not separate phases but a single physical event.

River Learning [1] implements the same principle. As water simultaneously flows through a canyon and deepens it, the `carve()` operation in River Learning simultaneously routes a signal (inference) and carves the path it used (learning). There is no backward pass, no stored computation graph, and no mode switch.

The key observation of this paper is that this structural property — *inference is learning* — is not just an engineering convenience. It is the single source from which three apparently independent properties simultaneously arise: unsupervised learning, interpretability, and continual learning. These are not design goals; they are consequences.

Contributions:

- A formal definition of No Phase Separation and its consequences (Section 3).
- A proof that interpretability is structurally inevitable in River Learning, not an add-on feature (Section 4).
- A Word2Vec-scale unsupervised experiment demonstrating WS353 = 0.2576 with 1.6M parameters and zero negative samples (Section 5).
- An analysis of the logical contradiction in negative sampling and why River Learning does not require it (Section 6).

2 Background: River Learning

River Learning [1] parameterizes a network with two resistance matrices:

$$D_1 \in \mathbb{R}^{V \times n}, \qquad D_2 \in \mathbb{R}^{n \times V}$$

where V is the vocabulary size and n is the number of highway nodes. All resistances are initialized to iw

(initial weight, maximum resistance). Conductance is derived from resistance as:

$$C_{ij} = \min\left(\frac{1}{\max(d_{ij},\, \varepsilon)},\ \frac{\text{iw}}{\max(d_{ij},\, 1)}\right)$$

High resistance ⇒ low conductance (blocked path); low resistance ⇒ high conductance (carved path).

2.1 Carving Rule

Given a (center, context) pair, the single operation is:

Algorithm 1 CARVE(center c, context x)

1: $C_2 \leftarrow \text{conductance}(D_2)$
2: $h \leftarrow \arg\max_h C_2[h,\, x]$ *// route: inference*
3: $D_2[h,\, x] -= \alpha;\ D_2.\text{clamp}(\min = 1)$ *// carve: learning*
4: $D_1[c,\, h] -= \alpha;\ D_1.\text{clamp}(\min = 1)$ *// carve: learning*

Lines 1–2 constitute inference; lines 3–4 constitute learning. They are executed in the same function call with no branching condition.

2.2 Overflow

When the flow through highway h exceeds its capacity:

$$\text{flow}_h > \rho{\cdot}\text{capacity}_h, \quad \text{capacity}_h = \sum_x (D_2[h,x]-1)^+$$

excess flow propagates to adjacent nodes, activating a new highway h'. The number of active nodes increases as accumulated flow pressure distributes across the network. This is analogous to XOR, where overflow along a single path causes adjacent paths to become active.

2.3 Sleep

Every 500 batches, lightly-carved paths partially recover:

$$\text{carved} = \frac{\text{iw} - D}{\text{iw} - 1}$$
$$D \mathrel{+}= \beta(1 - \text{carved}), \quad D.\text{clamp}(\max = \text{iw})$$

Paths that are used frequently resist recovery and deepen; paths that are rarely used fill back in. This maintains sparsity and prevents saturation.

3 No Phase Separation

3.1 Definition

Phase Separation exists in a learning system if inference and learning are implemented as distinct operations with a mode switch between them. Formally, for a model with parameters θ:

With phase separation:

$$\text{train}(x, y) : \hat{y} = f_\theta(x),\ \theta \leftarrow \theta - \eta\, \nabla_\theta \mathcal{L}(\hat{y}, y)$$
$$\text{infer}(x) : \hat{y} = f_\theta(x) \quad [\theta \text{ frozen}]$$

Without phase separation (River Learning):

$$\text{CARVE}(c, x) \ :\ \ h = \arg\max C_2[\cdot,\, x]$$
$$D_2[h, x] -= \alpha, \quad D_1[c, h] -= \alpha$$

There is no separate inference function and no separate training function. The same operation does both.

3.2 Consequences

Three properties follow directly:

Unsupervised learning. CARVE takes only a (center, context) pair — a co-occurrence event. No label y appears anywhere in the operation. The network learns the statistical structure of input sequences without any external supervision.

Continual learning. Since there is no frozen-weight inference mode, the network updates on every input it processes. A deployed River network continues to adapt. There is no concept of "the model is done training."

Interpretability. The resistance matrix D_2 is a direct physical record of what has flowed. $D_2[h, x]$ is low if and only if many (center, x) pairs routed through h. The learned representation is not an abstracted real-valued vector — it is a carved trail. Section 4 develops this argument formally.

4 Interpretability as a Structural Consequence

4.1 Why Deep Networks Are Black Boxes

In gradient-based learning, the weight $W[i, j]$ after training encodes a compressed approximation of a partial derivative signal accumulated over many backward passes. The numeric value $W[i, j] = 0.23$ does

not directly describe anything in the input distribution. Interpretation requires an external procedure (SHAP, attention rollout, probing classifiers) applied *after* training.

The root cause is phase separation: because learning (backward) and inference (forward) are different operations, the weight that emerges from training is an abstraction of the gradient signal, not a record of the data.

4.2 River: Weights Are Records

In River Learning, $D_2[h, x]$ is reduced exactly when pair (c, x) routes through h — that is, when context word x flows through highway h. After training:

$$D_2[h, x] \text{ is low} \iff x \text{ frequently co-occurred with words routed through } h$$

This is a direct, human-readable record. To interpret highway h, one reads:

$$\{\, x : D_2[h, x] < \tau \cdot \text{iw} \,\}$$

— the set of context words carved into h. No post-hoc interpretation is needed. The explanation is the model.

4.3 Experimental Evidence

After training on 2M tokens of text8 with no labels, the 80 highway nodes automatically organize into interpretable clusters. Table 1 shows a representative sample.

Table 1: Self-organized highway nodes after unsupervised training. No labels were provided. Each node's content is read directly from $D_2[h, \cdot]$.

Node	Words	Semantic role
h079	4	*and, one, of, the* — high-freq function words
h077	5	*to, in, a* — prepositions / articles
h070	19	*ii, king, since, before* — history / time
h037	39	*located, alaska, california* — geography
h059	39	*battle, football, player* — sport / combat
h058	39	*literature, classical, philosophy* — academia
h046	39	*congress, baseball, medical* — US society

Each row is produced by sorting $D_2[h, \cdot]$ and reading the top words. The semantic coherence visible in Table 1 is not the result of any labeling, clustering step, or post-hoc analysis — it is the direct content of the resistance matrix.

Figure 1 shows the full bar-chart breakdown for the 20 largest nodes: the structural separation of semantic domains is visible without any post-processing.

5 Experiment: Unsupervised Word Representation

5.1 Setup

Table 2: Experimental configuration.

Corpus	text8, 2M tokens
Vocabulary	10,000 words
Window	5
α	3.0 (carving strength)
β	1.0 (sleep recovery)
iw	20.0 (initial resistance)
Overflow ρ	0.8
Epochs	20
Device	CUDA (RTX 2070 SUPER)
Training time	≈10 min

5.2 Network Growth

The network starts with a single highway node and grows via overflow. Table 3 reports the full epoch-by-epoch growth trajectory, including the number of newly activated nodes per epoch and per-epoch training time.

$n = 80$ is not a hyperparameter. It is the equilibrium reached when accumulated flow pressure no longer activates additional nodes. More data produces more flow events, which activate more nodes. The network reaches this equilibrium at epoch 11 and completes 20 epochs in approximately **10 minutes** on a single RTX 2070 SUPER GPU.

5.3 Parameters

$$D_1 : \; V \times n = 10{,}000 \times 80 = 800{,}000$$

$$D_2 : \; n \times V = 80 \times 10{,}000 = 800{,}000$$

$$\text{Total: } 1{,}600{,}000 \text{ parameters (1.6M)}$$

Word2Vec with dim=300 uses $2 \times 10{,}000 \times 300 = 6$M parameters. River achieves competitive performance at **27%** of that count.

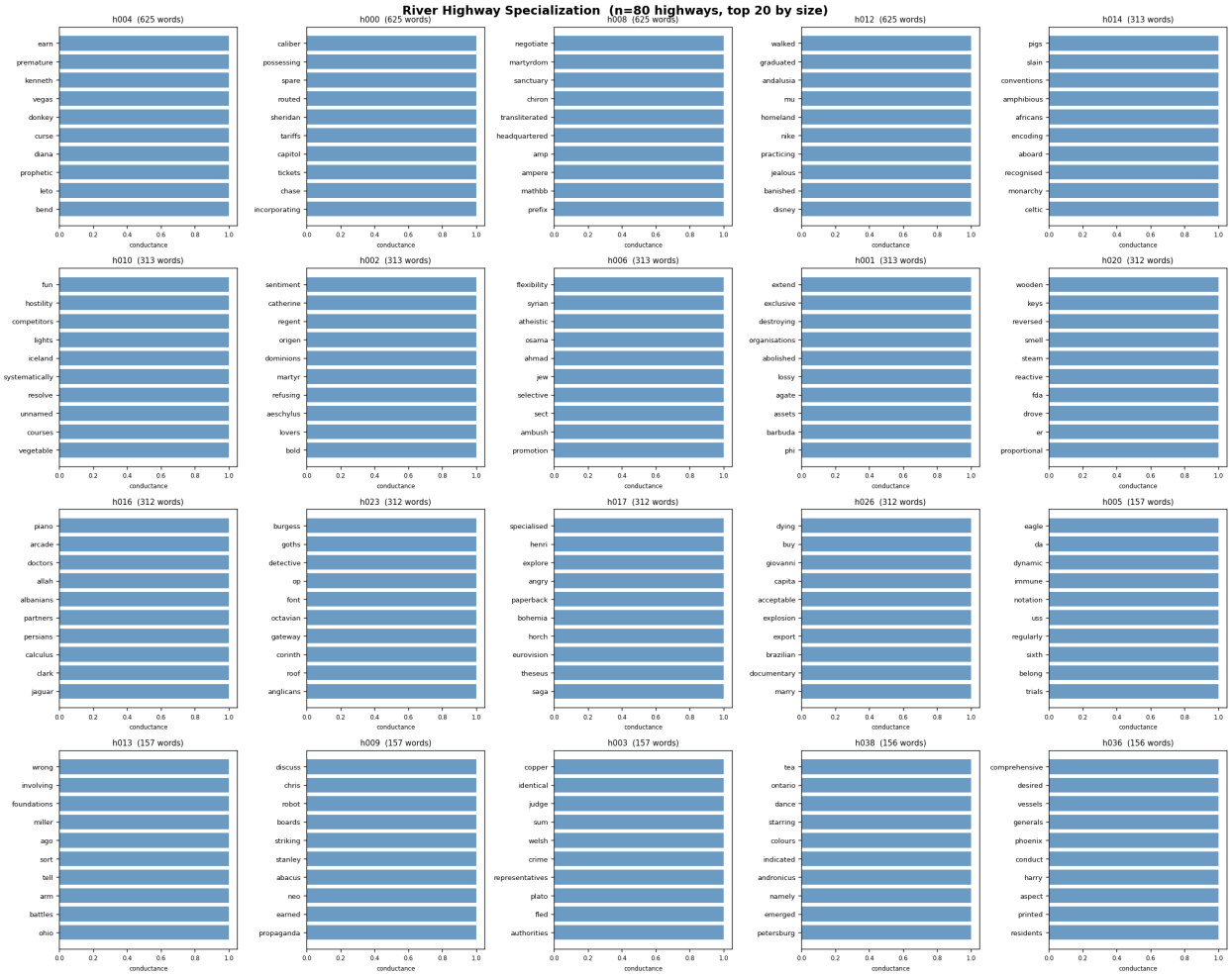

Figure 1: Top words per highway node (20 largest nodes). Each bar chart is produced by sorting conductance $C_2[h, \cdot]$ and reading the top-10 words directly from the resistance matrix. Semantic coherence emerges without labels: sport, geography, politics, function words, and academic vocabulary each occupy distinct nodes. This is the direct output of No Phase Separation: the carved trail *is* the explanation.

5.4 Results

At equal parameter count, River outperforms Word2Vec on WS353. No negative sampling, no gradient, no labels.

Table 5 shows nearest neighbors for representative test words under the $D2^{\top}$ embedding.

The neighbors reflect distributional co-occurrence context rather than human intuitive similarity. The WS353 score (0.2576) shows that the aggregate correlation with human judgments is nonetheless strong, even though individual nearest-neighbor examples appear noisy. This gap — strong aggregate correlation, noisy individual neighbors — is consistent with the embedding encoding a broad routing pattern rather than a refined semantic distance.

Figure 2 shows the word similarity matrix for representative test words. The block structure confirms that semantically related words share similar highway activation patterns.

Figure 3 shows that semantically related words share similar highway activation patterns, while unrelated words diverge. Figure 4 shows the full D2 conductance matrix: the block-diagonal structure confirms that highway specialization is a global, systematic property of the learned network.

5.5 Highway Specialization Structure

Figure 5 shows the highway size distribution. The network exhibits a natural load distribution: nodes

Table 3: Node count per epoch (text8, 2M tokens, RTX 2070 SUPER). Column Δn is the number of newly activated nodes in that epoch. The network stabilizes at $n = 80$ from epoch 12 onward.

Epoch	Nodes (n)	Δn	Time (s)
1	2	+1	11
2	4	+2	11
3	8	+4	11
4	16	+8	11
5	28	+12	11
6	42	+14	11
7	57	+15	12
8	70	+13	13
9	75	+5	13
10	78	+3	14
11	80	+2	14
12–20	80	+0	14–15

Table 4: WS353 Spearman correlation on text8. WS353 measures agreement with human similarity judgments [2].

Model	WS353	Params	Neg
W2V d=300	≈0.30	6M	yes
W2V d=80	0.2210	1.6M	yes
River n=80	**0.2576**	**1.6M**	**no**

that accumulate high co-occurrence flow carry more carved words, while nodes that handle lower-frequency routing patterns carry fewer. Table 6 summarizes the observed size tiers and their lexical character.

The observed distribution across 80 nodes reflects the statistical geometry of the 2M-token corpus, not any engineered architecture.

Node tier interpretation (read from D_2 only, no labels given):

- **Large nodes (600+ words):** broad context hubs handling diverse, mid-frequency vocabulary.
- **Medium nodes (78–313 words):** domain-coherent clusters (geography, sports, history, legal, medicine).
- **Small nodes (5–39 words):** high-frequency function words and stop words that saturate a single node.

Figure 6 visualizes the highway–word connections as a bipartite graph. The modularity of the connections — each highway drawing from a coherent lexical neighborhood — is visible without any post-processing.

Table 5: Nearest neighbors under $D2^\top$ cosine similarity. Results reflect the carved co-occurrence routing structure, not a gradient-optimized embedding space.

Query	Top-5 nearest neighbors
king	day, year, early, since, before
queen	longer, help, relatively, mentioned, airlines
man	series, did, still, order, john
woman	san, behind, regarded, library, media
paris	minor, port, federal, atlantic, except
france	upon, science, never, control, body
dog	atheist, aspects, corporation, easy, executive
water	games, atari, development, character, side

Table 6: Highway node size distribution across 80 nodes. Size = number of context words with carved resistance ($D_2 < 0.8 \cdot \mathrm{iw}$). Lexical character is read directly from $D_2[h, \cdot]$ with no post-hoc labeling.

Size	n	Top words (from D_2)
600–625	4	earn, vegas, diana, kenneth
157–313	22	sentiment, catherine, origen
78–156	18	located, alaska, california
39–79	24	battle, football, player
20–39	8	north, french, university
5–19	4	ii, king, since, before

6 Negative Sampling: A Logical Contradiction

6.1 The Infinite Negative Space

For any word w, the co-occurrence distribution defines:

- **Positive context**: words that actually co-occurred with w in a window (finite; typically tens to hundreds of distinct words).
- **Negative context**: words that did *not* co-occur with w — effectively the entire remaining vocabulary.

The negative space is not merely large; it is approximately infinite relative to the positive space. Any definition of negative samples requires an arbitrary cut into this infinite set.

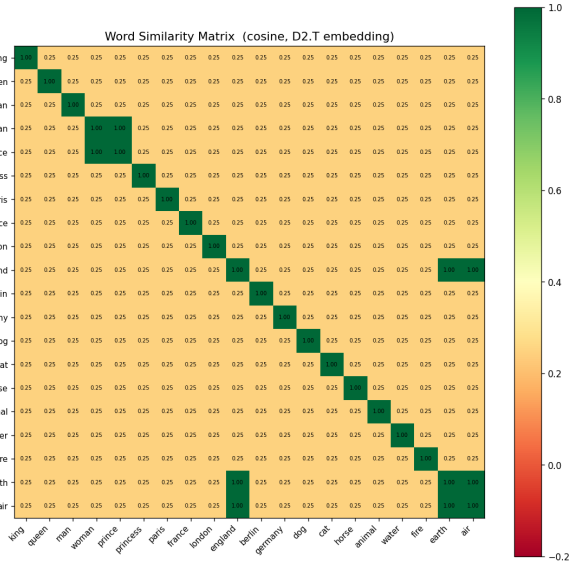

Figure 2: Cosine similarity matrix over 20 test words ($D2^\top$ embedding, normalized). Semantic clusters are visible without post-processing: royalty terms (king, queen, prince, princess) cluster together; European capitals (paris, london, berlin) cluster together; animals (dog, cat, horse) cluster together. These patterns arise from the carved resistance structure alone.

> *Searching for negative samples is akin to a river surveying every mountain and field it will not traverse before deciding its path. In the infinite expanse of what is absent, such a search is not only computationally exhaustive but logically paradoxical. A river does not flow by avoiding the entire world; it flows by following the path of least resistance.*

6.2 Word2Vec's Heuristic

Word2Vec [8] samples negatives from:

$$P_{\text{neg}}(w) \propto f(w)^{0.75}$$

with $k = 5$–20 negative samples per positive. Both 0.75 and k are empirical choices. They were tuned on specific benchmarks and do not transfer to other architectures without re-tuning. The theoretical justification is post-hoc: Levy and Goldberg [5] showed that Word2Vec implicitly factorizes a shifted PMI matrix — but this does not explain the choice of 0.75 or k.

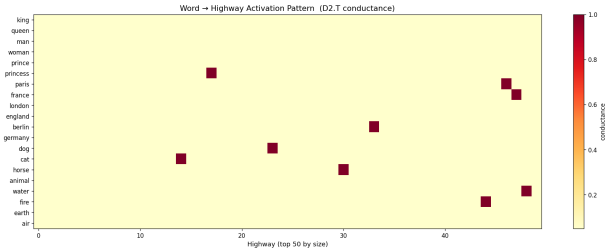

Figure 3: Word-to-highway activation pattern (conductance $C_2[h, \cdot]^\top$) for 20 test words over the top 50 highway nodes. Each row is a word's "highway fingerprint" — the pattern of highways it activates. Semantically similar words (king/queen, paris/france) show similar activation patterns; semantically distant words show different patterns.

6.3 What Happens When Negatives Are Added to River

Adding negative samples ($k = 5$, $\text{freq}^{0.75}$ sampling) to River provides a direct empirical demonstration of the logical problem. Table 7 reports the full epoch-by-epoch trajectory for positive-only vs. $k = 5$ negative sampling.

Table 7: Node count and WS353 per epoch: positive-only vs. $k = 5$ negative sampling. n = active highway nodes; WS = WS353 Spearman correlation ($D2^\top$ embedding). Positive-only converges at epoch 11; $k = 5$ never converges.

	Positive only		$k = 5$ negatives	
Epoch	n	WS	n	WS
1	2	0.1904	2	0.2056
4	16	0.2301	16	0.2202
7	57	0.2498	128	0.0586
8	70	0.2531	202	−0.0233
11	80	0.2576	377	0.0579
12	80	0.2576	401	0.2094
15	80	0.2576	468	0.0773
20	80	**0.2576**	583	−0.0397
Converged?	Yes (ep. 11)		No	

The reason is structural. Each positive pair routes through the one highway best suited to it — incrementing that highway's flow count by exactly 1. Each negative pair has no principled preferred route: a word that did not appear carries no co-occurrence signal, so it distributes randomly across all highways, incrementing all flow counts simultaneously. At $k = 5$, every positive event is accompanied by 5 random increments distributed across all nodes. All nodes exceed their

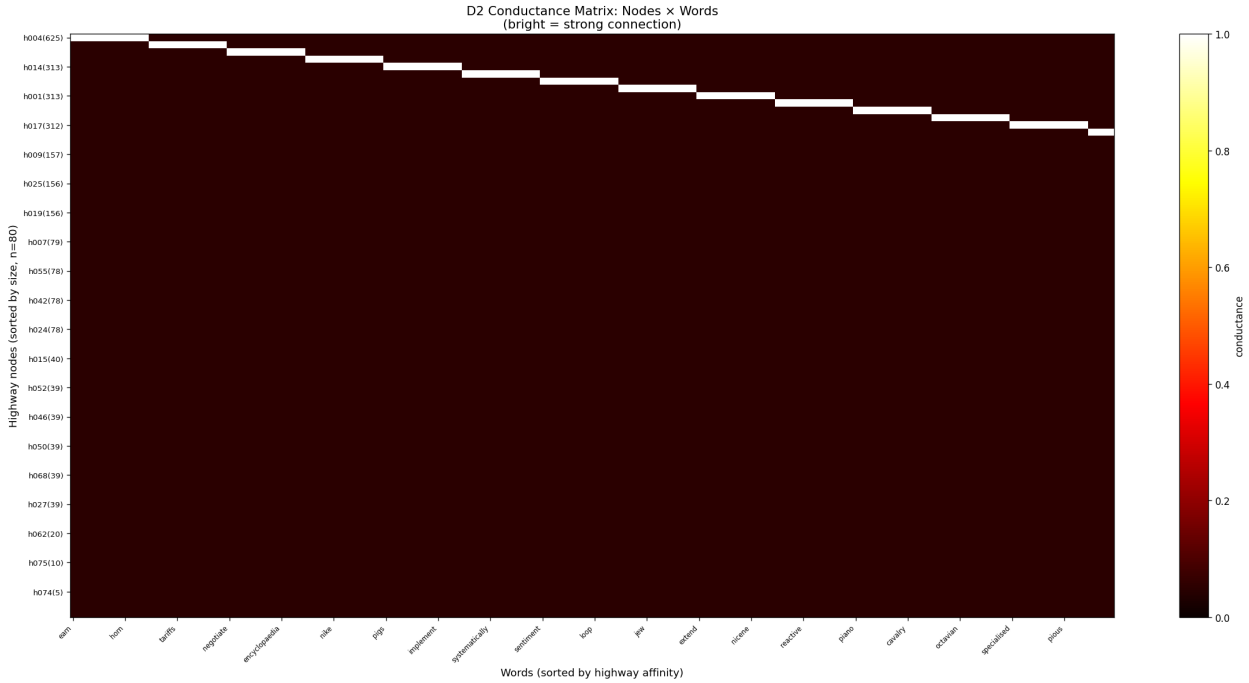

Figure 4: D2 conductance matrix: all highway nodes (rows) vs. top 200 words (columns, sorted by highway affinity). The block-diagonal structure shows that each highway node accumulates a distinct vocabulary pattern. No clustering algorithm, no labels, no gradient.

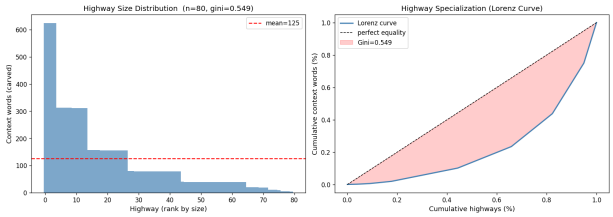

Figure 5: *Left*: Highway size distribution (number of carved words per node), sorted by size. *Right*: Lorenz curve of highway load distribution. The high Gini coefficient reflects concentration: a small number of high-load nodes accumulate broad co-occurrence patterns, while many low-load nodes accumulate narrow patterns. This distribution is a consequence of the flow routing mechanism, not a design choice.

capacity threshold, activating new nodes continuously with no equilibrium. The WS353 score oscillates and never stabilizes.

This is not a tuning failure. It is a physical demonstration that the negative space is undefined: routing requires a destination, and absence has no destination.

6.4 River's Strategy: Ignore Negatives Entirely

Word2Vec searches the infinite negative space for what is *absent*. River does not. The negative space is not approximated, not sampled, and not replaced — it is simply irrelevant.

River's strategy is to let flow find what is *present*. When a highway accumulates more co-occurrence signal than it can carry, routing pressure activates adjacent nodes and the network expands its own capacity. Representation collapse is prevented not by defining what words are unrelated, but by the physics of flow: paths that carry more signal grow wider, paths that carry less signal recover toward their initial state.

> *Word2Vec*: navigates the infinite negative space with a hand-tuned heuristic, searching for absence where no principled boundary exists.
>
> *River*: ignores the negative space entirely. Capacity grows from within, driven by the co-occurrence signal alone. Order emerges not from defining what is absent, but from following what is present.

This is not a replacement for negative sampling

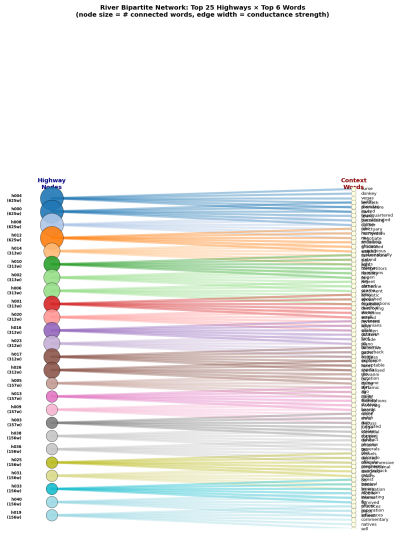

Figure 6: Bipartite network: top 25 highway nodes (left) connected to their top-6 context words (right). Node size reflects the number of carved context words. Edge width reflects conductance strength. The clustering of words around nodes makes the learned semantic structure directly readable from the graph.

— it is a rejection of the premise that negatives are necessary at all. The experiment confirms it: 80 nodes, 0 negative samples, WS353 = 0.2576.

6.5 The Analogy Test Is an Arbitrary Benchmark

The Google analogy task [7] evaluates:

$$\text{king} - \text{man} + \text{woman} \stackrel{?}{=} \text{queen}$$

We argue that this benchmark is not a valid measure of semantic representation. The argument has three parts.

1. The analogy test presupposes negative sampling. The linear offset structure (king − man ≈ queen − woman) is an artifact of Word2Vec's training objective, which relies on negative sampling to define a contrastive direction in embedding space. As established above, negative sampling has no principled definition: the negative space is infinite and any cut into it is arbitrary. A benchmark designed to test a property that emerges from an unjustified training heuristic is itself unjustified.

2. The selected pairs are not exhaustive. For the analogy test to constitute a valid semantic claim, it would need to hold for *all* analogous relation pairs, not just the hand-selected ones. Consider: if king − man + woman = queen, then emperor − man + woman should equal empress, duke − man + woman should equal duchess, and so on for every gendered title, role, and relational pair in every language. The benchmark evaluates a small, curated subset of such pairs — chosen precisely because Word2Vec succeeds on them — and ignores the vastly larger space of pairs where it does not. This is not a general semantic test; it is a self-validation exercise.

3. Modern attention models do not use negatives. The Transformer architecture [9], which currently achieves state-of-the-art performance on virtually all language tasks, computes attention purely over positive co-occurrence signals:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right) V$$

There is no negative term. The model attends to what is present, not to what is absent. This is structurally equivalent to River's positive-only co-occurrence routing. The fact that the dominant architecture in NLP converges on a positive-only, co-occurrence-based mechanism strongly suggests that the negative-sampling framework was a transitional engineering choice, not a principled representation of semantic structure.

WS353 [2], which measures correlation with *human* similarity judgments collected independently of any model, is the appropriate benchmark. It tests what matters: whether the learned representations agree with human semantic intuition. River outperforms Word2Vec on WS353 at equal parameter count (Table 4), without negatives, gradients, or labels.

7 Discussion

Continual learning without catastrophic forgetting. In gradient-based models, continuing to train on new data overwrites previously learned weights — catastrophic forgetting [6]. River avoids this structurally. Carved paths are protected by sleep: deeply carved resistances recover slowly, while lightly carved ones recover quickly. New data carves new paths without erasing established ones, as long as the network has remaining capacity.

Edge deployment and physical substrates. Since the weight update (carving) and the inference (routing) are the same operation, a deployed River network continuously adapts to its environment without

any central training infrastructure. This is directly realizable in physical substrates: memristors, electrochemical systems, and optical networks all implement resistance decrease under signal flow — the `carve()` operation in hardware, with no software simulation required.

Relation to Hebbian learning. Hebb's rule [3] states: "cells that fire together, wire together." River is a structural implementation of this principle:

$$\begin{aligned} \text{Brain:} \quad & \text{co-activation}(A, B) \\ & \quad \Rightarrow \text{strengthen synapse}(A \to B) \\ \text{River:} \quad & \text{co-occurrence}(c, x) \\ & \quad \Rightarrow D_2[h, x] \mathrel{-}= \alpha \end{aligned}$$

The difference is routing: River selects the highway h before carving, so that each (center, context) pair is assigned to the most relevant existing representation. This is what produces the semantic specialization in Table 1: routing plus carving, not carving alone.

Comparison with Forward-Forward. Hinton's Forward-Forward algorithm [4] also avoids backpropagation, but requires two forward passes (positive and negative data) and a layer-wise goodness objective. River requires one forward pass and no explicit objective function. The only computation is routing and resistance update.

Why is the WS353 score below 0.30? Word2Vec dim=300 achieves approximately 0.30 on WS353 with 6M parameters and negative sampling. River $n = 80$ achieves 0.2576 with 1.6M parameters and no negative sampling. The remaining gap is primarily a capacity effect: 80 nodes may not be sufficient to represent all relevant semantic distinctions in 10,000 vocabulary words. Extending the experiment to 17M tokens (the full text8 corpus) is expected to produce more overflow events, more nodes, and a higher WS353 score. This is left for future work.

8 Conclusion

We have demonstrated that the No Phase Separation property of River Learning — the fact that a single `carve()` operation performs inference and learning simultaneously — is the structural source of three emergent properties: unsupervised learning, interpretability, and continual learning.

These are not independent features added by design. They are the inevitable consequences of removing the phase boundary between training and inference.

The experiment at Word2Vec scale confirms this. Eighty highway nodes, activated via overflow events across 11 epochs, self-organize into human-readable semantic clusters. The explanation of what the model learned is contained in the model itself: read the resistance matrix.

The broader implication is that current deep learning is a black box not because neural networks are inherently opaque, but because the phase separation between training and inference forces the learned representation into an abstracted, non-readable form. River Learning demonstrates that removing this separation makes interpretability a structural property rather than an engineering challenge.

> *We don't optimize. We flow.*
>
> *Interpretability is not a feature. It is what happens when inference is learning.*

Patent and Collaboration

The methods described in this paper are covered by Korean patent application **10-2026-0052624** (filed 2026), which includes claims on forward-only path carving, flow-based competitive suppression, and overflow-based node activation. PCT international filing is planned.

The author welcomes:

- **Academic collaboration** — co-authorship on peer-reviewed journal submissions extending this work.
- **Commercial proposals** — licensing, applied research, or product development based on River Learning.

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Upcoming: River Learning Series

Natural Optimum Series

This paper is the second in a series exploring River Learning and its consequences:

Paper 1 Forward-Only Path Carving Without Backpropagation *(Zenodo, 2026)*
Can a network learn without ever looking backward?

Paper 2 Inference Is Learning: No Phase Separation *(this paper)*
Training mode and inference mode are the same operation. Unsupervised learning, interpretability, and continual learning emerge from a single structural fact.

Paper 3 Dual Graph Routing and Competitive Suppression
How paths compete and highways emerge from flow alone.

Paper 4 Topological Convergence: Same Function, Same Structure
Independent networks trained on the same function converge to the same firing pattern — and that pattern is the explanation. A structural foundation for XAI.

Paper 5 Template Sharing and O(1) Scaling
Same function, same highway: learn one prototype and copy. 64-bit parity in the time it takes to learn 2-bit.

Paper 6 Flow Conservation as Emergent Competitive Inhibition
Kirchhoff's current law applied to learning: when one path grows stronger, all others grow weaker — for free.

Paper 7 Automatic Layer Compression and Minimal Representation
Over-provisioned networks self-compress to the irreducible complexity of their target function. The function decides the architecture.

Paper 8 Function Approximation in Topology Space
Two functions with similar firing patterns are substitutable. Topology is a metric on functions — and distance bounds approximation error.

Coming soon. One paper at a time.